Final Exam Prepare - 2/10 exam

Q1 - Chap 8(ntp) + Chap 4(syslog) all theory but maybe will come out command

1. TWO advantages of using NTP (Page 3)
Automatic updates — No flashing 12:00 lights
Flexible & Accurate — You control how clocks are updated

2. Stratum 0 (Page 10)
- High-precision timekeeping devices.
- Very accurate, such as GPS clock

3. Stratum 1
- Devices directly connected to Stratum 0.
- Primary time servers

4. Stratum 2 / Stratum 2 and Lower (Page 11)
- Synchronized to stratum 1 servers
- Responds to time requests from other NTP clients

5. Stratum 16 (Page 9)
- Stratum 16 is the lowest stratum level.
- It indicates that a device is unsynchronized.

6. Role of NTP client and NTP server
NTP client
- Requests time updates from NTP server
NTP server
- Responds to time requests from other NTP clients
- Does not modify their own time

7. Cisco IOS command for router to function as NTP client
- ntp server ip-address

8. Verify NTP configuration

- Before NTP is configured, `show clock detail` shows **"Time source is user configuration"**.
- After NTP is configured, `show clock detail` shows **"Time source is NTP"**.

9. 自己看回去slides的page12 - 16来复习command part, 可能会出

Chap 4

10. syslog standard addresses three primary components (Page 4)
- Event message format
- Event message transmission
- Event message handling

11. THREE primary functions of Syslog logging (Page 6)
- The ability to gather logging information for monitoring and troubleshooting
- The ability to select the type of logging information that is captured
- The ability to specify the destinations of captured syslog messages

10. Syslog severity names (Page 10)

Severity Name	Severity Level	Explanation
Emergency	Level 0	System Unusable
Alert	Level 1	Immediate Action Needed
Critical	Level 2	Critical Condition
Error	Level 3	Error Condition
Warning	Level 4	Warning Condition
Notification	Level 5	Normal, but Significant Condition
Informational	Level 6	Informational Message
Debugging	Level 7	Debugging Message

11. Popular destinations for Syslog messages（Page 8)

- Logging buffer
- Console line
- Terminal line

12.Syslog message facilities on Cisco IOS routers (Page 11)

- SYS — operating system
- IP security (IPsec)
- Interface IP (IF)

13. By default, the format of syslog messages on the Cisco IOS Software is as follows:
%facility-severity-MNEMONIC: description

For example, sample output on a Cisco switch for an EtherChannel link changing state to up is:

%LINK-3-UPDOWN: Interface Port-channel1, changed state to up

Two type of Ans(看question要什么，如果没有明确要什么，就2选一):
(i)
Here the facility is LINK and the severity level is 3, with a MNEMONIC of UPDOWN.

(ii)
facility = LINK
severity = 3（Error Condition）
MNEMONIC = UPDOWN
description = Interface Port-channel1, changed state to up

14. Configure Syslog Timestamp (Page 14)

Command: service timestamps log datetime

Q2 - Chap 10(Network Automation) + Chap 11(Network Troubleshooting)

Chap10

1. THREE common data formats & it characteristics(Page 6-19)

JavaScript Object Notation (JSON)
- It uses braces { } to hold objects and square brackets [ ] hold arrays.
- Its data is written as key/value pairs.

eXtensible Markup Language (XML)
- It is self-descriptive. It encloses data within a related set of tags: <tag>data</tag>.
- Unlike HTML, XML uses no predefined tags

YAML Ain't Markup Language (YAML)
- It is like JSON and is considered a superset of JSON.
- It has a minimalist format making it easy to read and write.



2. JSON error identification + solution (Page 13-14)

答题 format

(i) Identify TWO lines

写法：The errors are on Line X and Line Y. + explain what is wrong

(ii) Explain TWO solutions

写法（每个 solution 对应一个 error）：

Solution 1 (for Line X): [explain what is wrong and how to fix it]

Solution 2 (for Line Y): [explain what is wrong and how to fix it]

#用 slides 的 JSON syntax rules 来解释，或者你自己解释都可以

All Rules

- JSON uses braces { } to hold objects.
- JSON uses square brackets [ ] to hold arrays.
- In JSON, keys must be strings within double quotation marks "".
- In JSON, values must be a valid JSON data type, which can be a string, number, array, Boolean, null, or another object.
- In JSON, keys and values are separated by a colon.
- In JSON, multiple key/value pairs within an object are separated by commas.
- In JSON, each value in the array is separated by a comma.

六、常见 JSON error 类型（考试一定会中一个）

Error 类型	例子	怎样 fix
Missing comma	`"name": "abc"` 后面还有 key	加 `,`
Wrong bracket	`{ }` 用在 array；`[ ]` 用在 object	根据内容改 `{ }` 或 `[ ]`
Spelling error	`"addre"`, `"notmask"`	改成 `"address"`, `"netmask"`
Missing quotes	`name: "abc"`	key 加 `" "`
Missing colon	`"name" "abc"`	加 `:`
Trailing comma	最后一个 key/value pair 后面有 `,`	删掉 `,`

3. RESTful API — features (Page 27)
Client-Server
- The client handles the front end and the server handles the back end.

Cacheable
- Clients can cache responses to improve performance.

6. RESTful API — Anatomy of a RESTful Request (Page 31-32)
These are the different parts of the API request
- API Server

- Resources

- Query — THREE subcomponents of a query
  - Format
  - Key
  - Parameters

7. RESTful API Applications - A RESTful API request can also be made in other ways.（Page 33）
- Postman
- Python

8. These are some characteristics of the network that administrators benefit from automating: (Page 42)
- Software and version control
- Protocol configurations

9. Configuration Management Tools (Page 43)
- Ansible
- Chef

10. Protocols/tool used to support network automation (Page 41)
- REST
- JSON
- Puppet
- chef

Chap 11

1. Two Network Topology Diagrams (Page 5)
- Physical topology
- Logical topology

3. Step Troubleshooting Process (Page 14) - seven step:
1. Define the Problem
2. Gather Information
3. Analyze Information
4. Eliminate Possible Causes
5. Propose Hypothesis
6. Test Hypothesis
7. Solve the Problem

5. Structured Troubleshooting Methods (Page 18)

Bottom-Up
- Good method to use when the problem is suspected to be a physical one

Top-Down
- Use this method for simpler problems

Divide-and-Conquer
- Start at a middle layer (i.e, Layer 3) and test in both directions from that layer.

Follow-the-Path
- Used to discover the actual traffic path from source to destination to reduce the scope of troubleshooting.

6. Software Troubleshooting Tools (Page 21)
- Network Management System Tools
- Baselining Tools

7. Hardware Troubleshooting Tools (Page 23)
- Digital Multimeters
- Cable Testers

8. Guidelines of selecting a Troubleshooting Method (Page 19)
会问根据什么problem选什么method去解决

Define problem - Determine the type of problem. - Analyze the symptoms. - Apply previous experiences.	Type of problem	Troubleshooting Method
	Software-oriented	Top-Down Method
	Hardware / Cable-oriented or Complex	Bottom-Up Method
	New Problem	Bottom-Up Method
	Experienced Before	Divide-and-Conquer Method

Q3 - Chap 6(WLAN Concepts) + Chap 9(VPN and IPsec) all theory

Chap 6

2. Types of Wireless Networks (Page 5)
Wireless Personal-Area Network (WPAN)
Wireless LAN (WLAN)
Wireless MAN (WMAN)
Wireless WAN (WWAN)

3. Wireless Technologies (Page 7-8)
- Bluetooth
- WiMAX

5. Wireless Router Functions (Page 16)
Access point — To provide wireless access
Switch — To interconnect wired devices
Router — To provide a default gateway to other networks and the Internet

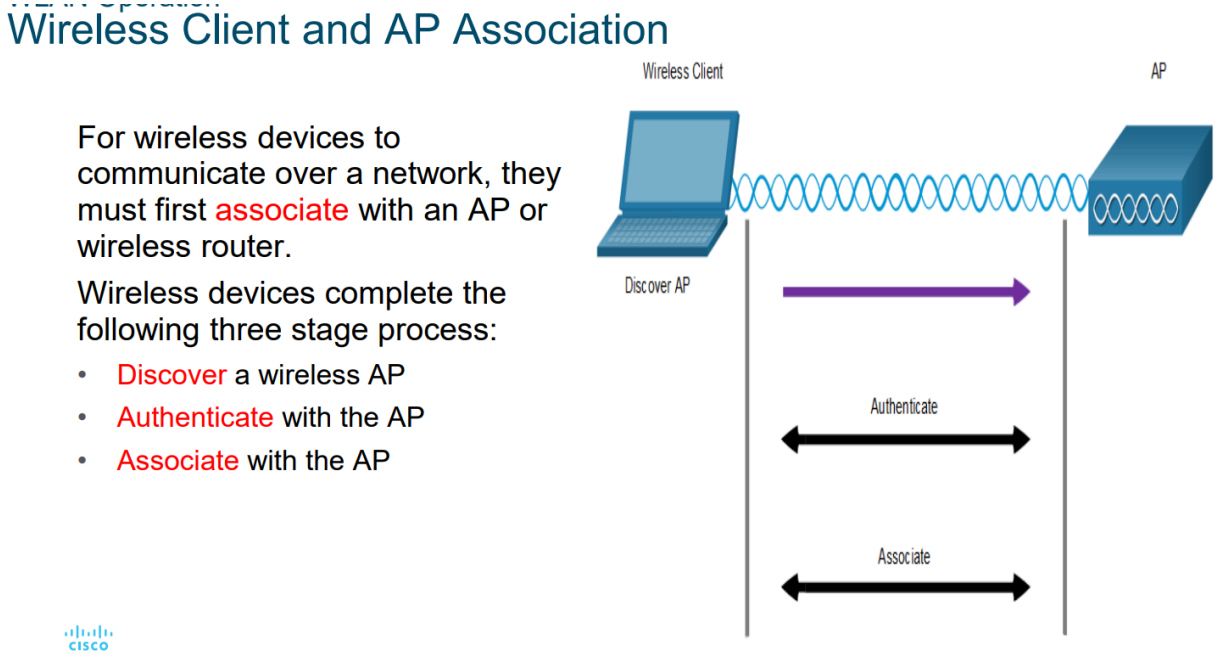
6. AP Categories (Page18)
Autonomous APs
- Standalone devices configured through CLI or GUI.
Controller-based APs
- Also known as lightweight APs (LAPs) and use LWAPP to communicate with a WLC.

9. BSS & ESS (Page 23)
Basic Service Set (BSS)
- Uses single AP to interconnect all associated wireless clients.
Extended Service Set (ESS)
- A union of two or more BSSs interconnected by a wired distribution

system.

10. Wireless Client and AP Association (Page 26-27)

#Three-stage process to associate with AP also at 12.3



To achieve successful association, a wireless client and an AP must agree on specific parameters:

- SSID
- Password
- Network mode
- Security mode

11. Wireless Security Overview — Four Threats (Page 42)

- Interception of data
- Denial of Service (DoS) Attacks
- Rogue APs

12. All step setup (maybe will come out)

12.1 FIVE steps to set up wireless network

1. Configure the wireless router's basic settings
2. Set the SSID
3. Set the password / pre-shared key
4. Select the network mode and security mode
5. Configure channel settings

12.2 THREE steps to access GUI config mode

1. Connect to the wireless router
2. Open a web browser and enter the router's default IP address
3. Log in with the default username and password

12.3 Three-stage process to associate with AP

1. Discover a wireless AP
2. Authenticate with the AP
3. Associate with the AP

12.4 CSMA/CA process

1. Listen to the channel to see if it is idle.
2. Sends a Ready to Send (RTS) message to the AP to request dedicated access.
3. Receives a Clear to Send (CTS) message from the AP granting access to send.
4. Transmits the data.
5. Acknowledges all transmissions.

Chap 9

1. Definition of VPN（Page 4)

VPNs create end-to-end private network connections.

2.Two benefits of using a VPN (Page 5)

Cost Savings — Organizations can use VPNs to reduce their connectivity costs

Security — Encryption and authentication protocols protect data from unauthorized access.

3. Site-to-Site VPN (Page 6 / 12)

- A site-to-site VPN is terminated on VPN gateways.
- VPN traffic is only encrypted between the gateways.

4. Remote-Access VPN (Page 7)

A remote-access VPN is dynamically created to establish a secure connection between a client and a VPN terminating device.

5. Two types of Remote-Access VPN

Clientless VPN connection

- The connection is secured using a web browser SSL connection.

Client-based VPN connection

- VPN client software

6. Enterprise VPNs vs Service Provider VPNs (Page 8)

	Enterprise VPNs	**Service Provider VPNs**
Created and managed by	The enterprise	The provider network
Technology used	IPsec and SSL VPNs	MPLS at Layer 2 or 3

7. IPsec vs SSL Remote Access Deployments

Features	IPsec	SSL
Applications supported	Extensive — All IP-based applications	Limited — Only web-based applications and file sharing
Connection complexity	Medium — Requires VPN client installed on a host	Low — Requires web browser on a host

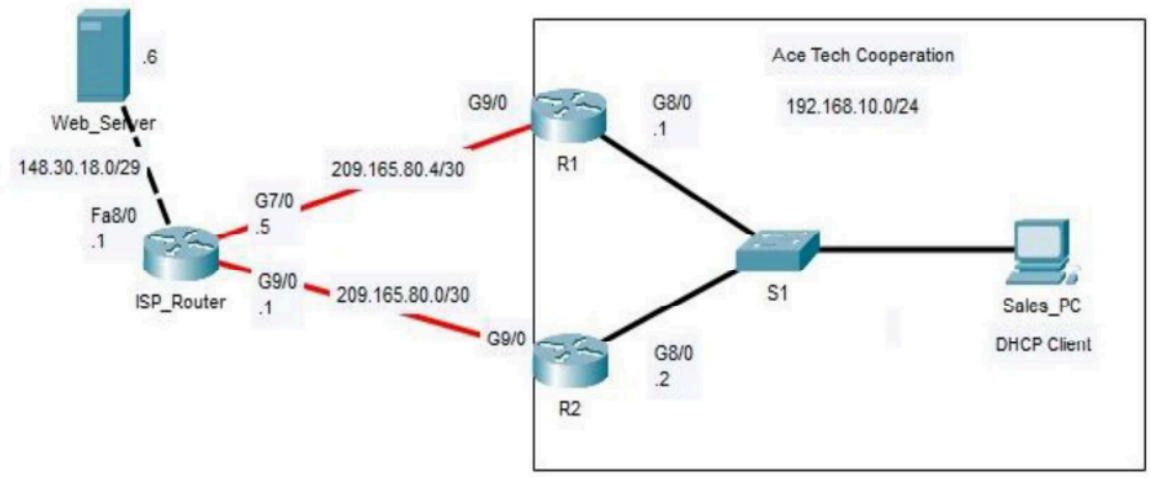
Q4 - Chap 5(DHCPv4) + Chap 3(SNMP)

1. DHCPv4 Configuration Example

Exp Format:

```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
R1(config)# ip dhcp excluded-address 192.168.10.254
R1(config)# ip dhcp pool LAN-POOL-1
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 192.168.11.5
R1(dhcp-config)# domain-name example.com
R1(dhcp-config)# end
```

202509 - Q4 (iv)



(iii) Identify the router in Figure 2 that acts as the DHCP Server allowing Sales_PC to obtain an IPv4 address. (1 mark)

这边的说的是Ace Tech Cooperation 的地方

(iv) Provide **FOUR (4)** Cisco IOS configuration commands to be applied on the router identified in Question 4 a) (iii), with the following requirements:

- Exclude the first two IP addresses from the 192.168.10.0/24.
- Create a DHCP pool named LAN_10.
- Configure the pool to support the 192.168.10.0/24 network.
- The default gateway is 192.168.10.1. (10 marks)

Ans:

ip dhcp excluded-address 192.168.10.1 192.168.10.2 (First two IP address so, 0-2 => 1-2)
ip dhcp pool LAN_10 (create pool and named it)
network 192.168.10.0 255.255.255.0 (Ace Tech Cooperation的ip address跟subnet mask)
default-router 192.168.10.1 (default gateway is 192.168.10.1)

2025 - Q4 (i)

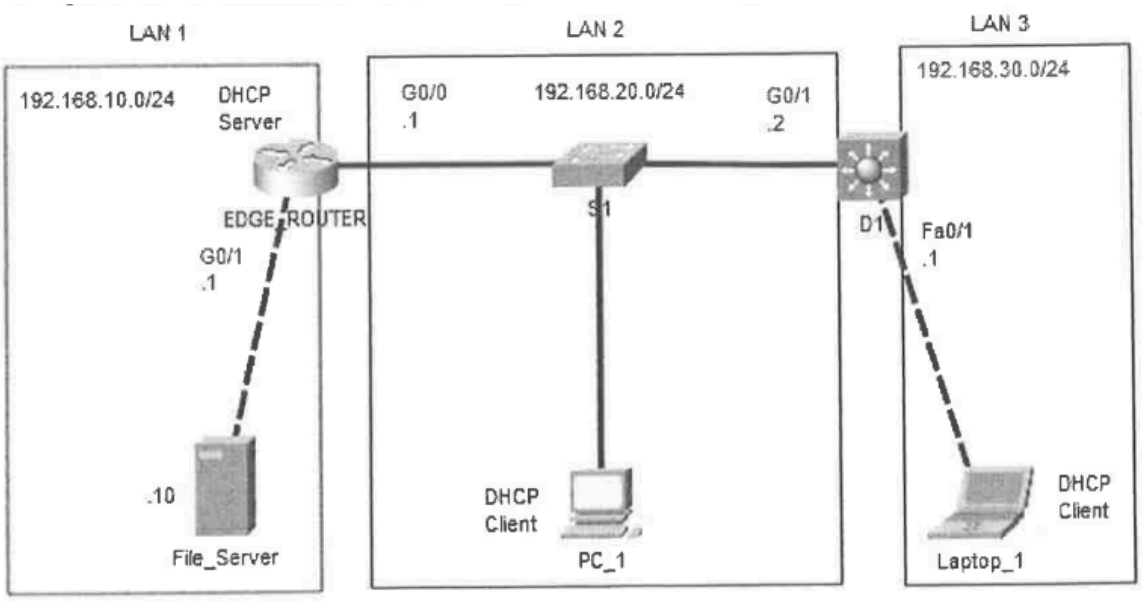


Figure 4-1: Network topology diagram of Ace Tech.

(i) Provide **FIVE (5)** Internetwork Operating System (IOS) configuration commands to be applied on EDGE_ROUTER based on the following requirements:
- Exclude the first two IP addresses from LAN 3.
- Create a DHCP pool for LAN 3 named as LAN-3-Addresses.
- The default gateway for LAN 3 is 192.168.30.1.
- The Domain Name System (DNS) server for LAN 3 is the File_Server, which has the IP address 192.168.10.10. (12 marks)

Ans:
ip dhcp excluded-address 192.168.30.1 192.168.30.2
ip dhcp pool LAN-3-Addresses
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 192.168.10.10

其他会根据DHCP出的小题

a) Ace Tech Cooperation is a new organisation. The company is using two routers, R1 and R2. Both routers are connected to an Internet Service Provider (ISP). R1 and R2 use fibre optic cables to connect to the ISP router (ISP_Router). The G9/0 interfaces on R1 and R2 use public Internet Protocol version 4 (IPv4) addresses allocated by the ISP. R1 and R2 of Ace Tech Cooperation are connected to a Local Area Network (LAN) with a network address of 192.168.10.0/24. The ISP_Router is also connected to a Web_Server. A Sales_PC from Ace Tech Cooperation is able to display a website hosted on the Web_Server via the ISP_Router. The network topology diagram is shown in Figure 2.

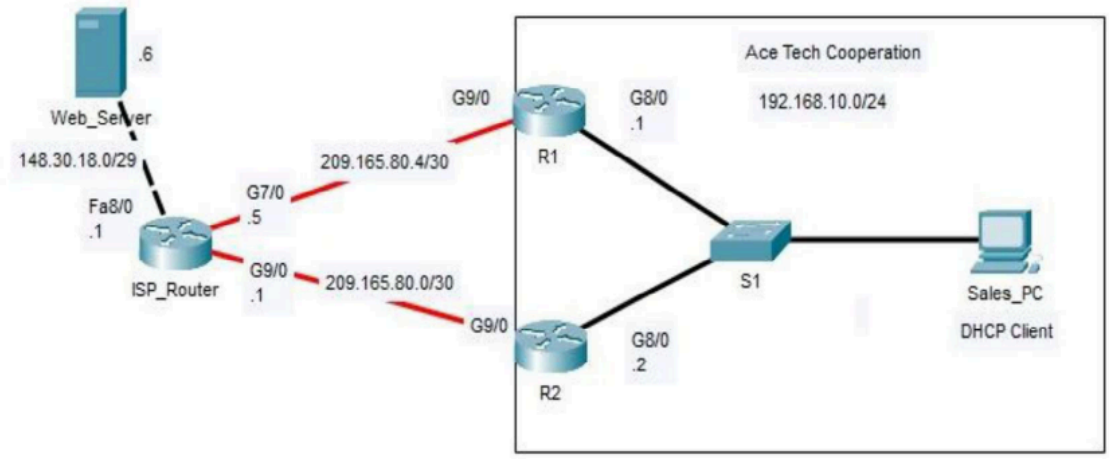


Figure 2: Network topology diagram of Ace Tech Cooperation.

(i) Which router in Figure 2 is referred as the Dynamic Host Configuration Protocol (DHCP) server that provides public IP addresses to Ace Tech Cooperation? (1 mark)

(ii) Write **TWO (2)** Cisco IOS configuration commands that are applied to R1 in order for R1 to obtained a public IP address. (2 marks)

(iii) Identify the router in Figure 2 that acts as the DHCP Server allowing Sales_PC to obtain an IPv4 address. (1 mark)

Ans:

(i) ISP_Router

(ii) R1(config)# interface G9/0
R1(config-if)# ip address dhcp

(iii) R1（OR R2）

2. Write TWO (2) Cisco IOS configuration commands that are applied to R1 in order for R1 to obtain a public IP address. (类似这样的)
这种第一种普通的就一样
interface g0/1
ip address dhcp

(ii) Configure DHCP relay on D1 to enable Laptop_1 to successfully acquire IP parameters from the DHCP server. Provide TWO (2) IOS configuration commands based on this requirement. (3 marks)
或者是有讲DHCP delay的，就用
interface fa0/1
ip helper-address 192.168.20.1

通常都是问这些之类的小题，所以**theory part**我直接丢了

Chap 3 theory

1. **SNMP function**

- monitor network performance and device status

2. SNMP system consists of three elements

SNMP manager

- Monitors and control network devices

F - send SNMP requests and receive responses

SNMP management agents

- Software on managed device that collects & reports info to NMS

F - Responds to manager requests and sends traps / alerts

Management Information Base (MIB)

- Database containing information about managed device

F - Store information that the manager and agent access

SNMP advantages

- Easy network monitoring
- Detects problems quickly.

SNMP operation

operation	description
get	SNMP manager retrieves info from an agent
set	SNMP manager changes configuration on an agent
trap	The agent sends an alert to the manager without being asked.

SNMP polling scenario (manager to agent)

b) Figure 3 shows a Simple Network Management Protocol (SNMP) setup between an SNMP Manager and an SNMP Agent.

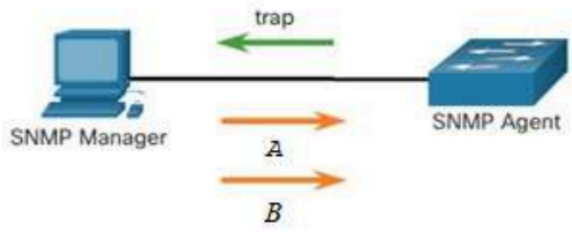


Figure 3: An SNMP Manager and an SNMP Agent.

Discuss **TWO (2)** actions, labelled *A* and *B*, that are performed by the SNMP Manager as indicated in Figure 3. (8 marks)

[Total: 25 marks]

A = get-request
The SNMP manager retrieves a value from a specific variable on the SNMP agent.
It is used to collect information from the SNMP agent.

B = set-request
The SNMP manager stores a value in a specific variable on the SNMP agent.
It is used to change configurations on the SNMP agent.

（补充）trap
Trap is an unsolicited message sent from the SNMP agent to the SNMP manager, alerting the manager to a condition or event on the network.